

Python: exercises on iteration

This notebook was generated in **solutions** mode.

Exercise 1: Print Numbers from 1 to N

Exercise Description

Write a function that prints all numbers from 1 to N using a for loop and range.

Your solution

```
def print_numbers(n):  
    """Print all numbers from 1 to n."""  
    for i in range(1, n + 1):  
        print(i)
```

Test your solution

```
# Test case 1: Test with small number
print_numbers(5)
```

Debug your solution

The following cells contain code to generate links to Python Tutor for debugging your code when running the tests. **You need to first run the cell containing your solution code before running the cell containing the Python Tutor link.**

```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pyth
```

Exercise 2: Sum of First N Natural Numbers

Exercise Description

Write a function to calculate the sum of the first N natural numbers ($1 + 2 + 3 + \dots + N$) using a while loop.

Your solution

```
def sum_natural_numbers(n):
    """Calculate sum of first n natural numbers."""
    i = 1
    total = 0
    while i <= n:
```

```
        total += i
    i += 1
    return total
```

Test your solution

```
# Test case 2: Test with another number
result = sum_natural_numbers(3)
expected = 1 + 2 + 3 # 6
assert result == expected, f"Expected {expected}, got {result}"
print(f"✓ sum_natural_numbers(3) = {result}")
```

Test your solution

```
# Test case: Example from solution
result = sum_natural_numbers(5)
print(f"Sum of first 5 natural numbers: {result}") # 15
```

Debug your solution

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```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pyth
```

Exercise 3: Print Even Numbers from 2 to 20

Exercise Description

Write a function that prints all even numbers between 2 and 20 using a for loop and range.

Your solution

```
def print_even_numbers():  
    """Print even numbers from 2 to 20."""  
    for i in range(2, 21, 2):  
        print(i)
```

Test your solution

```
# Test case 2: Verify range and step  
# Expected: prints 2, 4, 6, 8, 10, 12, 14, 16, 18, 20 (each on new line)  
print_even_numbers()
```

Debug your solution

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```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pyth
```

Exercise 4: Count Down from N to 1 (If Even)

Exercise Description

Write a function to print numbers from N to 1 using a while loop, but only if N is even.

Your solution

```
def countdown_if_even(n):  
    """Count down from n to 1, but only if n is even."""  
    if n % 2 == 0:  
        while n > 0:  
            print(n)  
            n -= 1
```

Test your solution

```
# Test case 2: Test with odd number (should print nothing)  
countdown_if_even(3)
```

Test your solution

```
# Test case 2 (assert)  
countdown_if_even(6) # Prints 6, 5, 4, 3, 2, 1
```

Debug your solution

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```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pyth
```

Exercise 5: Multiplication Table of 3

Exercise Description

Write a function that prints the multiplication table of 3 (from 3x1 to 3x10) using a for loop.

Your solution

```
def multiplication_table_3():  
    """Print multiplication table of 3."""  
    for i in range(1, 11):  
        print(f"3 x {i} = {3 * i}")
```

Test your solution

```
# Test case 2: Verify format  
# Expected: prints 3 x 1 = 3, 3 x 2 = 6, ..., 3 x 10 = 30
```

```
multiplication_table_3()
```

Debug your solution

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```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pyth
```

Exercise 6: Sum of First N Even Numbers

Exercise Description

Write a function to calculate the sum of the first N even numbers (2 + 4 + 6 + ...) using a while loop.

Your solution

```
def sum_first_n_even(n):  
    """Calculate sum of first n even numbers."""  
    i = 2  
    total = 0  
    count = 0  
    while count < n:  
        total += i
```

```
        i += 2
        count += 1
    return total
```

Test your solution

```
# Test case 2: Test with another number
result = sum_first_n_even(5)
expected = 2 + 4 + 6 + 8 + 10 # 30
assert result == expected, f"Expected {expected}, got {result}"
print(f"✓ sum_first_n_even(5) = {result}")
```

Debug your solution

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```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pyth
```

Exercise 7: Numbers Divisible by 5 (1 to 50)

Exercise Description

Write a function that prints all numbers divisible by 5 between 1 and 50 using a for loop and range.

Your solution

```
def print_divisible_by_5():  
    """Print numbers divisible by 5 between 1 and 50."""  
    for i in range(5, 51, 5):  
        print(i)
```

Test your solution

```
# Test case 2: Verify range  
# Expected: prints 5, 10, 15, 20, 25, 30, 35, 40, 45, 50  
print_divisible_by_5()
```

Debug your solution

The following cells contain code to generate links to Python Tutor for debugging your code when running the tests. **You need to first run the cell containing your solution code before running the cell containing the Python Tutor link.**

```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pyth
```

Exercise 8: Factorial of 5

Exercise Description

Write a function to find the factorial of 5 using a while loop.

Your solution

```
def factorial_5():  
    """Calculate factorial of 5."""  
    i = 5  
    factorial = 1  
    while i > 0:  
        factorial *= i  
        i -= 1  
    return factorial
```

Test your solution

```
# Test case 2: Verify result  
result = factorial_5()  
assert result == 120, f"Expected 120, got {result}"  
print(f"✓ Factorial of 5 is {result}")
```

Debug your solution

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```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pyth
```

Exercise 9: Sum of Odd Numbers (1 to 20)

Exercise Description

Write a function to calculate the sum of all odd numbers between 1 and 20 using a for loop.

Your solution

```
def sum_odd_numbers():  
    """Calculate sum of odd numbers between 1 and 20."""  
    total = 0  
    for i in range(1, 21, 2):  
        total += i  
    return total
```

Test your solution

```
# Test case 2: Verify result  
result = sum_odd_numbers()  
assert result == 100, f"Expected 100, got {result}"  
print(f"✓ Sum of odd numbers 1-20 is {result}")
```

Test your solution

```
# Test case: Example from solution  
result = sum_odd_numbers()
```

```
print(f"Sum of odd numbers 1-20: {result}") # 100
```

Debug your solution

The following cells contain code to generate links to Python Tutor for debugging your code when running the tests. **You need to first run the cell containing your solution code before running the cell containing the Python Tutor link.**

```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pyth
```

Exercise 10: Numbers Divisible by K (1 to N)

Exercise Description

Write a function to print numbers between 1 and N that are divisible by K using a for loop and range.

Your solution

```
def print_divisible_by_k(n, k):  
    """Print numbers between 1 and n that are divisible by k."""  
    for i in range(k, n + 1, k):  
        print(i)
```

Test your solution

```
# Test case 2: Test with n=20, k=4
# Expected: prints 4, 8, 12, 16, 20
print_divisible_by_k(20, 4)
```

Test your solution

```
# Test case: Example from solution
print_divisible_by_k(20, 3) # Prints 3, 6, 9, 12, 15, 18
```

Debug your solution

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```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pyth
```

Exercise 11: Count Digits in a Number

Exercise Description

Write a function that counts how many digits are in a positive integer.

Your solution

```
def count_digits(num):  
    """Return the number of digits in positive integer num."""  
    if num == 0:  
        return 1  
    count = 0  
    while num > 0:  
        num //= 10  
        count += 1  
    return count
```

Test your solution

```
assert count_digits(123456) == 6  
assert count_digits(7) == 1  
assert count_digits(10) == 2  
print("✓ count_digits tests passed")
```

Debug your solution

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```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pyth
```

Exercise 12: Cubes from 1 to 5

Exercise Description

Write a function that returns a list with the cubes of numbers from 1 to 5 using a for loop.

Your solution

```
def cubes_1_to_5():  
    result = []  
    for i in range(1, 6):  
        result.append(i**3)  
    return result
```

Test your solution

```
assert cubes_1_to_5() == [1, 8, 27, 64, 125]  
print("✓ cubes_1_to_5 tests passed")
```

Debug your solution

The following cells contain code to generate links to Python Tutor for debugging your code when running the tests. **You need to first run the cell containing your solution code before running the cell containing the Python Tutor link.**

```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pyth
```

Exercise 13: Product of Numbers from 1 to N

Exercise Description

Write a function that returns the product of numbers from 1 to N using a for loop.

Your solution

```
def product_1_to_n(n):  
    product = 1  
    for i in range(1, n + 1):  
        product *= i  
    return product
```

Test your solution

```
assert product_1_to_n(1) == 1  
assert product_1_to_n(3) == 6  
assert product_1_to_n(5) == 120  
print("✓ product_1_to_n tests passed")
```

Debug your solution

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```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pyth
```

Exercise 14: Odd Numbers from 1 to 15

Exercise Description

Write a function that prints all odd numbers from 1 to 15 using a while loop.

Your solution

```
def odd_1_to_15():  
    i = 1  
    while i <= 15:  
        print(i)  
        i += 2
```

Test your solution

```
import io  
import sys  
  
# Capture the printed output  
captured_output = io.StringIO()  
sys.stdout = captured_output  
  
odd_1_to_15()
```

```
# Restore normal output
sys.stdout = sys.__stdout__

# Get the output and convert to list of integers for testing
output_lines = captured_output.getvalue().strip().split('\n')
result = [int(line.strip()) for line in output_lines if line.strip()]

assert result == [1, 3, 5, 7, 9, 11, 13, 15]
print("✓ odd_1_to_15 tests passed")
print(f"Generated odd numbers: {result}")
```

Debug your solution

The following cells contain code to generate links to Python Tutor for debugging your code when running the tests. **You need to first run the cell containing your solution code before running the cell containing the Python Tutor link.**

```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pyth
```

Exercise 15: Print Numbers from 5 to 1

Exercise Description

Write a function that prints numbers from 5 to 1 using a for loop with range.

Your solution

```
def countdown_5_to_1():  
    for i in range(5, 0, -1):  
        print(i)
```

Test your solution

```
import io  
import sys  
  
# Capture the printed output  
captured_output = io.StringIO()  
sys.stdout = captured_output  
  
countdown_5_to_1()  
  
# Restore normal output  
sys.stdout = sys.__stdout__  
  
# Get the output and convert to list of integers for testing  
output_lines = captured_output.getvalue().strip().split('\n')  
result = [int(line.strip()) for line in output_lines if line.strip()]  
  
assert result == [5, 4, 3, 2, 1]  
print("✓ countdown_5_to_1 tests passed")  
print(f"Generated countdown: {result}")
```

Debug your solution

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```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pyth
```

Exercise 16: Sum of the First 7 Odd Numbers

Exercise Description

Write a function to find the sum of the first 7 odd numbers using a while loop.

Your solution

```
def sum_first_7_odds():  
    i = 1  
    count = 0  
    total = 0  
    while count < 7:  
        total += i  
        i += 2  
        count += 1  
    return total
```

Test your solution

```
assert sum_first_7_odds() == 49
print("✓ sum_first_7_odds tests passed")
```

Debug your solution

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```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pyth
```

Exercise 17: Repeat String as Many Times as Its Length

Exercise Description

Write a function that takes a string and returns a concatenated string with the original string repeated as many times as its length. Use a loop and remember that `len(s)` gives you the length of string `s`.

Your solution

```
def repeat_string_len_times(s):
    result = ""

    for i in range(len(s)):
```

